

2025-2026 – Econ 0016 – International Macroeconomics

Chapter 8: The Twin Deficits: Fiscal Deficits and the Current Account

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0. Introduction

- ▶ An important question in macroeconomics is whether fiscal deficits cause current account deficits.
- ▶ There are 2 opposing views:
 1. tax-cut induced fiscal deficits lead to current account deficits (twin deficits);
 2. tax cuts have no effect on the current account (Ricardian equivalence).
- ▶ **Twin deficits:** A tax cut generates a fiscal deficit but increases household income and stimulates consumption spending, deteriorating the current account.
- ▶ **Ricardian equivalence:** A tax cut generates a fiscal deficit and hence public debt. In the future this debt must be repaid, which requires an increase in taxes. Households understand this and save the tax cut to pay the higher future taxes. Consumption is unchanged.
- ▶ This chapter analyzes conditions under which each of the two views are valid. In addition, it studies whether similar opposing views arise when the fiscal deficit is a consequence of changes in government consumption rather than reductions in taxes (or increases in government transfers).

1. An Open Economy with a Government Sector

- ▶ Take the two-period small open economy of Chapter 5;
- ▶ add a government that consumes goods (government spending), levies taxes, and issues debt.

1. An Open Economy with a Government Sector

The Government

- ▶ G_t = government purchases of goods and services in period $t = 1, 2$, which are exogenously given.
- ▶ T_t = lump-sum taxes in period $t = 1, 2$. The term lump-sum taxes refers to taxes that do not depend on any economic characteristic of the taxpayer, such as income, spending, or wealth.
- ▶ Lump-sum taxes are rarely seen in real-world applications, but they make the analysis simpler.
- ▶ Later in this chapter, we will look at distortionary taxation.
- ▶ B_{t-1}^g = bond holdings of the government at the start of period t for $t = 1, 2$.
- ▶ If $B_{t-1}^g < 0$, we say there is public debt outstanding.

1. An Open Economy with a Government Sector

The Government

- ▶ The government budget constraint in period 1:

$$G_1 + B_1^g - B_0^g = T_1 + r_0 B_0^g, \quad (1)$$

- ▶ The left-hand side represents outlays in period 1: purchases of goods, G_1 , and purchases of bonds, $B_1^g - B_0^g$.
- ▶ The right-hand side represents sources of funds in period 1: tax revenues, T_1 , and interest income on bonds, $r_0 B_0^g$.
- ▶ The government budget constraint in period 2:

$$G_2 + B_2^g - B_1^g = T_2 + r_1 B_1^g. \quad (2)$$

- ▶ At the end of period 2, the government cannot leave debts and does not want to leave assets (transversality condition). Thus

$$B_2^g = 0. \quad (3)$$

1. An Open Economy with a Government Sector

The Government

- ▶ Combining (1) to (3) to eliminate B_1^g and B_2^g , we obtain the intertemporal government budget constraint:

$$G_1 + \frac{G_2}{1 + r_1} = (1 + r_0)B_0^g + T_1 + \frac{T_2}{1 + r_1}. \quad (4)$$

- ▶ This equation says that the present discounted value of government consumption (the left-hand side) must be equal to the sum of initial asset holdings including interest and the present discounted value of tax revenue (the right-hand side).
- ▶ There exist many tax policies T_1 and T_2 that can finance a given path of government consumption, G_1 and G_2 . But given taxes in one period, taxes in the other period are pinned down.
- ▶ In particular, a tax cut in period 1 must be offset by a tax increase in period 2.

1. An Open Economy with a Government Sector

Firms

- ▶ As in Chapter 5, firms borrow in period 1 to invest in capital goods that become productive in period 2.

$$Q_2 = A_2 F(l_1),$$

where Q_2 denotes output in period 2, A_2 is an exogenous productivity factor, l_1 is investment made in period 1 that becomes productive in period 2 and $F(\cdot)$ is an increasing and concave production function.

- ▶ In period 2, the firm must repay the loan with interest. The firm chooses l_1 to maximize profits:

$$\Pi_2 = A_2 F(l_1) - (1 + r_1)l_1.$$

- ▶ First-order optimality condition is

$$A_2 F'(l_1) = 1 + r_1$$

which implies that investment is decreasing in r_1 and independent of taxes or government spending

$$l_1 = l(r_1). \quad (5)$$

1. An Open Economy with a Government Sector

Firms

- Using (5), we can express profits as

$$\Pi_2 = A_2 F(I(r_1)) - (1 + r_1)I(r_1).$$

- As in Chapter 5, we then have

$$\Pi_2 = \Pi(\underline{r_1}). \quad (6)$$

- Firms are assumed to be owned by households to whom they distribute profits in period 2.
- Income in period 1 is assumed to be given by Q_1 .

1. An Open Economy with a Government Sector

Households

$$\max_{C_1, C_2} \ln C_1 + \ln C_2 \quad (7)$$

subject to

$$C_1 + B_1^h - B_0^h = r_0 B_0^h + Q_1 - T_1 \quad (8)$$

$$C_2 + B_2^h - B_1^h = r_1 B_1^h + \Pi(r_1) - T_2, \quad (9)$$

where B_t^h denotes the bond holdings of the household at the end of period t , for $t = 0, 1, 2$.

- ▶ Income net of taxes is called disposable income, that is, $Q_1 - T_1$ and $\Pi(r_1) - T_2$.
- ▶ The transversality condition implies that

$$B_2^h = 0. \quad (10)$$

1. An Open Economy with a Government Sector

Households

- Combining equations (8) to (10) yields the household's intertemporal budget constraint

$$C_1 + \frac{C_2}{1+r_1} = (1+r_0)B_0^h + Q_1 - T_1 + \frac{\Pi(r_1) - T_2}{1+r_1} \quad (11)$$

- Solve (11) for C_2 :

$$C_2 = (1+r_1) \left((1+r_0)B_0^h + Q_1 - T_1 - C_1 \right) + \Pi(r_1) - T_2. \quad (12)$$

1. An Open Economy with a Government Sector

Households

- ▶ To find the optimal consumption path, use (12) to eliminate C_2 from the utility function (7).
- ▶ This operation gives

$$\ln C_1 + \ln \left((1 + r_1)(\bar{Y} - C_1) \right) \quad (13)$$

where

$$\bar{Y} = (1 + r_0)B_0^h + Q_1 - T_1 + \frac{\Pi(r_1) - T_2}{1 + r_1}$$

- ▶ The household chooses C_1 to maximize (13), taking as given $\bar{Y}$ and r_1 .

1. An Open Economy with a Government Sector

Households

- The resulting optimality condition is $C_1 = \frac{1}{2}\overline{Y}$ or

$$C_1 = \frac{1}{2} \left((1 + r_0)B_0^h + Q_1 - T_1 + \frac{\Pi(r_1) - T_2}{1 + r_1} \right). \quad (14)$$

- Note that C_1 depends only on the present discounted value of taxes, $T_1 + T_2/(1 + r_1)$.
- Using (14) to eliminate C_1 from (12) gives

$$C_2 = \frac{1 + r_1}{2} \left((1 + r_0)B_0^h + Q_1 - T_1 + \frac{\Pi(r_1) - T_2}{1 + r_1} \right), \quad (15)$$

which says that also C_2 depends only on the present discounted value of taxes, $T_1 + T_2/(1 + r_1)$

1. An Open Economy with a Government Sector

Equilibrium

- ▶ Because the economy is small and there is free capital mobility, the domestic interest rate, r_1 , equals the world interest rate, r^*

$$r_1 = r^* \quad (16)$$

- ▶ The country's net foreign asset position at the beginning of period 1, denoted B_0 , is given by the sum of private and public asset holdings; that is,

$$B_0 = B_0^h + B_0^g. \quad (17)$$

- ▶ Combining (4), (14), (16), and (17) yields the equilibrium level of consumption in period 1,

$$C_1 = \frac{1}{2} \left((1 + r_0)B_0 + Q_1 - G_1 + \frac{\Pi(r^*) - G_2}{1 + r^*} \right). \quad (18)$$

- ▶ A similar operation using (15) delivers the equilibrium level of consumption in period 2,

$$C_2 = \frac{1 + r^*}{2} \left((1 + r_0)B_0 + Q_1 - G_1 + \frac{\Pi(r^*) - G_2}{1 + r^*} \right). \quad (19)$$

1. An Open Economy with a Government Sector

Equilibrium

- The trade balance in period 1, denoted TB_1 , is

$$TB_1 = Q_1 - C_1 - G_1 - I_1.$$

- The current account in period 1, denoted CA_1 , is

$$CA_1 = TB_1 + r_0 B_0.$$

- Using (5) and (18) to eliminate I_1 and C_1 , we have

$$TB_1 = \frac{1}{2} \left(-(1 + r_0)B_0 + Q_1 - G_1 - \frac{\Pi(r^*) - G_2}{1 + r^*} \right) - I(r^*). \quad (20)$$

and

$$CA_1 = \frac{1}{2} \left(-(1 - r_0)B_0 + Q_1 - G_1 - \frac{\Pi(r^*) - G_2}{1 + r^*} \right) - I(r^*). \quad (21)$$

1. An Open Economy with a Government Sector

Equilibrium

- ▶ Neither T_1 nor T_2 appear in equations (18)-(21).
- ▶ $\rightsquigarrow$ given G_1 and G_2 , any combination of taxes T_1 and T_2 satisfying the government's intertemporal budget constraint (4) is associated with the same equilibrium levels of C , TB and CA .
- ▶ This result is known as *Ricardian equivalence*.
- ▶ Intuition:
 - × Suppose the government cuts taxes in period 1, and leaves G_1 and G_2 unchanged.
 - × Then the tax cut must be financed with public debt.
 - × Repaying the higher public debt requires increasing taxes in period 2.
 - × In period 1, Hh anticipate that the tax cut will require higher taxes in period 2.
 - × Consequently, instead of spending part of the tax cut on consumption goods, households keep consumption unchanged and save all of the tax cut.
- ▶ This intuition for why Ricardian equivalence holds in this economy is based on the idea that a tax cut (i.e., a reduction in government saving) is exactly offset by an increase in private saving.
- ▶ Let's show this result more formally.

2. Ricardian Equivalence

- ▶ **Government saving**, denoted S_1^g , is

$$S_1^g = r_0 B_0^g + T_1 - G_1.$$

- ▶ Given G_1 and $r_0 B_0^g$, any change in T_1 must be reflected one-for-one by a change in government saving; that is,

$$\Delta S_1^g = \Delta T_1. \quad (22)$$

- ▶ **Private saving**, denoted S_1^p , is

$$S_1^p = Q_1 + r_0 B_0^h - T_1 - C_1.$$

- ▶ By (18), for given G_1 and G_2 , C_1 depends only on the total net present value of taxes $T_1 + \frac{T_2}{1+r^*}$ (which has not changed).
- ▶ Thus, given $Q_1 + r_0 B_0^h$, any change in T_1 leads to a change in private saving of equal size but opposite sign; that is,

$$\Delta S_1^p = -\Delta T_1. \quad (23)$$

2. Ricardian Equivalence

- ▶ **National saving**, denoted S_1 , is $S_1 = S_1^p + S_1^g$.
- ▶ and changes in national saving hence are

$$\Delta S_1 = \Delta S_1^g + \Delta S_1^p = \Delta T_1 - \Delta T_1 = 0.$$

- ▶ This expression confirms the intuition we gave for why Ricardian equivalence holds in this economy.
- ▶ Given the path of government spending, a change in taxes causes changes in government and private saving that exactly offset each other, leaving national saving unchanged.
- ▶ With investment unaffected by the tax change, we have that the current account is also unchanged, that is, the model fails to predict twin deficits.
- ▶ Can the model predict twin deficits when fiscal deficits are caused by changes in G as opposed to changes in T ?

3. Government Spending and Twin Deficits

- ▶ A change in G_1 (that can be financed by an increase in taxes today or by an increase in debt), has the same effect as a change in the endowment but in the opposite direction.
- ▶ To see this, take a look at equilibrium conditions (18)-(21):

$$C_1 = \frac{1}{2} \left((1 + r_0)B_0 + Q_1 - G_1 + \frac{\Pi(r^*) - G_2}{1 + r^*} \right), \quad (18)$$

$$C_2 = \frac{1 + r^*}{2} \left((1 + r_0)B_0 + Q_1 - G_1 + \frac{\Pi(r^*) - G_2}{1 + r^*} \right), \quad (19)$$

$$TB_1 = \frac{1}{2} \left(-(1 + r_0)B_0 + Q_1 - G_1 - \frac{\Pi(r^*) - G_2}{1 + r^*} \right) - I(r^*), \quad (20)$$

$$CA_1 = \frac{1}{2} \left(-(1 + r_0)B_0 + Q_1 - G_1 - \frac{\Pi(r^*) - G_2}{1 + r^*} \right) - I(r^*). \quad (21)$$

- ▶ G_1 always appears in the form $Q_1 - G_1$.
- ▶ We can use what we know about effect of transitory and permanent endowment shocks.

3. Government Spending and Twin Deficits

A temporary increase in government spending

- ▶ A temporary increase in government spending reduces private consumption but by less than the change in government spending itself. (Partial crowding out of private consumption)

$$\Delta C_1 = -\frac{1}{2}\Delta G_1$$

- ▶ Because the world interest rate is unaffected, the change in government spending has no effect on investment:

$$\Delta I_1 = 0$$

- ▶ Hence overall effect on domestic absorption is positive:

$$\Delta(C_1 + I_1 + G_1) = -\frac{1}{2}\Delta G_1 + 0 + \Delta G_1 = \frac{1}{2}\Delta G_1 > 0$$

- ▶ Finally, by assumption, output in period 1 is fixed at Q_1 :

$$\Delta Q_1 = 0$$

3. Government Spending and Twin Deficits

A temporary increase in government spending

- Increase in G_1 leads to TB_1 and CA_1 deficits:

$$\Delta TB_1 = \Delta Q_1 - \Delta(C_1 + I_1 + G_1) = 0 - \frac{1}{2}\Delta G_1 = -\frac{1}{2}\Delta G_1 < 0.$$

$$\Delta CA_1 = \Delta TB_1 < 0$$

- What about twin deficits?
 - × If the increase in government spending is not fully financed by an increase in current taxes, the fiscal deficit in period 1 increases.
 - × In this case, the expansion in government spending causes an increase in both the fiscal and the current account deficits $\rightsquigarrow$ twin deficits.
 - × But if it is accompanied with a commensurate increase in current taxes ($\Delta T_1 = \Delta G_1$), then there is no twin deficit.

3. Government Spending and Twin Deficits

An expected future increase in government spending

- ▶ Experiment: In period 1 it is learned that G_2 will increase (a “news”).
- ▶ In equilibrium conditions (18)-(21), G_2 always appears together with period-2 income, $\Pi(r^*) - G_2$.
- ▶ Thus, an expected increase in G_2 is equivalent to an expected fall in income in that period and lowers C_1 :

$$\Delta C_1 = -\frac{1}{2(1+r^*)}\Delta G_2 < 0$$

- ▶ The fall in C_1 is smaller than the increase in G_2 because households smooth the adjustment over time.
- ▶ (20) and (21), the increase in G_2 leads to an improvement in the trade balance and the current account in period 1.

$$\Delta TB_1 = \Delta CA_1 = \frac{1}{2(1+r^*)}\Delta G_2 > 0.$$

- ▶ Twin Surpluses/Deficits? In general not, only if taxes move to make it so.

3. Government Spending and Twin Deficits

A permanent increase in government spending

- ▶ Experiment: $\Delta G_1 = \Delta G_2 = \Delta G > 0$
- ▶ The current account should not change much. The intuition is similar to that behind a permanent fall in the endowment.
- ▶ Formally,

$$\Delta CA_1 = \Delta TB_1 = \Delta Q_1 - \Delta(C_1 + I_1 + G_1) = -\frac{r^*}{2(1+r^*)}\Delta G,$$

which is a small number if r^* is small.

- ▶ It follows that permanent increases in government spending in general do not lead to twin deficits.
- ▶ In sum, the present model can capture the twin deficit hypothesis when changes in the fiscal deficit are caused by temporary changes in government spending, but in general cannot capture it when they are caused by permanent or future expected changes in government spending.

4. Failure of Ricardian Equivalence: Tax Cuts and Twin Deficits

- ▶ The conclusion that tax cuts don't cause twin deficits is a consequence of Ricardian equivalence.
- ▶ The assumptions under which Ricardian equivalence obtains, however, are special:
 - × households face no borrowing constraints,
 - × households expect tax cuts to be followed by higher future taxes
 - × all taxes are lump sum.
- ▶ When relaxing any of these three assumptions, as we will show next, tax cuts can lead to twin deficits.
- ▶ Main takeaway: Ricardian equivalence is a fragile result.

4. Failure of Ricardian Equivalence: Tax Cuts and Twin Deficits

Borrowing Constraints

- ▶ An example of a simple borrowing (or liquidity) constraint is that households can save but that they cannot borrow:

$$B_1^h \geq 0$$

- ▶ Absent of this constraint, Hh would choose point B on the next slide figure.
- ▶ Assume that this constraint is binding for the initial level of taxes (point A) and also after the tax cut (point A').

4. Failure of Ricardian Equivalence: Tax Cuts and Twin Deficits

Borrowing Constraints



4. Failure of Ricardian Equivalence: Tax Cuts and Twin Deficits

Borrowing Constraints

- We have

$$\Delta C_1 = -\Delta T_1.$$

$\rightsquigarrow$ Ricardian equivalence fails.

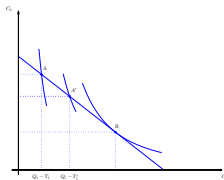
- Neither firms nor the government are liquidity constrained $\rightsquigarrow I_1$ and G_1 are unaffected by the tax cut.
- Q_1 is given $\rightsquigarrow TB_1$ and CA_1 deteriorate by the same amount as the increase in consumption:

$$\Delta TB_1 = \Delta CA_1 = \Delta T_1 < 0.$$

- This is because the tax cut causes a reduction in public saving (or an increase in the fiscal deficit) in period 1:

$$\Delta S_1^g = \Delta T_1 < 0.$$

- It follows that in the presence of borrowing constraints, tax cuts generate twin deficits.



4. Failure of Ricardian Equivalence: Tax Cuts and Twin Deficits

Intergenerational Effects

- ▶ A second reason why Ricardian equivalence could fail is that those who benefit from a tax cut are not the ones that pay for the tax increase later.
- ▶ Consider an endowment economy in which households live for only one period.
- ▶ Budget constraint of the generation alive in period 1:

$$C_1 + T_1 = Q_1$$

- ▶ Budget constraint of the generation alive in period 2:

$$C_2 + T_2 = Q_2.$$

- ▶ Tax cut in period 1 paid for with tax increase in period 2:

$$\Delta C_1 = -\Delta T_1 > 0 \text{ and } \Delta C_2 = -\Delta T_2 < 0.$$

- ▶ Neither firms nor the government are short-lived $\rightsquigarrow I_1$ and G_1 are unaffected by the tax cut.
- ▶ Therefore trade balance and the current account in period 1 decline one-for-one with the decline in taxes $\rightsquigarrow$ twin deficits.

4. Failure of Ricardian Equivalence: Tax Cuts and Twin Deficits

Distortionary Taxation

- ▶ Finally, Ricardian equivalence may also break down if taxes are distortionary rather than lump sum.
- ▶ Example: proportional tax on consumption, τ_1 and τ_2 .
- ▶ After-tax cost of consumption becomes $(1 + \tau_1)C_1$ and $(1 + \tau_2)C_2$.
- ▶ Effect of tax cut in period 1, a decline in τ_1 , financed with an increase in τ_2 :
 - × C_1 becomes cheaper relative to C_2 ,
 - × households consume more in period 1 and less in period 2.
 - × Thus, unlike changes in lump-sum taxes, changes in distortionary taxes can have real effects.
- ▶ Let's derive this breakdown of Ricardian equivalence more formally.

4. Failure of Ricardian Equivalence: Tax Cuts and Twin Deficits

Distortionary Taxation: Household

- Household budget constraints in periods 1 and 2:

$$(1 + \tau_1)C_1 + B_1^h - B_0^h = r_0 B_0^h + Q_1, \quad (24)$$

$$(1 + \tau_2)C_2 + B_2^h - B_1^h = r_1 B_1^h + \Pi(r_1). \quad (25)$$

- Combining these two budget constraints and the transversality condition (10) yields the intertemporal budget constraint:

$$(1 + \tau_1)C_1 + \frac{(1 + \tau_2)C_2}{1 + r_1} = (1 + r_0)B_0^h + Q_1 + \frac{\Pi(r_1)}{1 + r_1} \quad (26)$$

4. Failure of Ricardian Equivalence: Tax Cuts and Twin Deficits

Distortionary Taxation: Household

- Household chooses C_1 and C_2 to

$$\max_{C_1, C_2} \ln C_1 + \ln C_2 \quad (7)$$

subject to (26), taking as given τ_1 , τ_2 , Q_1 , $(1 + r_0)B_0^h$, and r_1 .

- Solve (26) for C_2 and then eliminate C_2 from utility function (7), to obtain

$$\max_{C_1} \left\{ \ln C_1 + \ln \left(\left(\frac{1 + r_1}{1 + \tau_2} \right) (\bar{Y} - (1 + \tau_1)C_1) \right) \right\}$$

with $\bar{Y} = (1 + r_0)B_0^h + Q_1 + \frac{\Pi(r_1)}{1 + r_1}$.

4. Failure of Ricardian Equivalence: Tax Cuts and Twin Deficits

Distortionary Taxation: Household

- First order condition is

$$\frac{C_2}{C_1} = \left(\frac{1 + \tau_1}{1 + \tau_2} \right) (1 + r_1). \quad (27)$$

- We see that cuts in τ_1 financed by increases in τ_2 will make the household consume relatively more in period 1 and relatively less in period 2.
- A consumption tax distorts the intertemporal relative price of consumption.
- The household perceives that one unit of consumption in period 1 costs $\left(\frac{1 + \tau_1}{1 + \tau_2} \right) (1 + r_1)$ units of consumption in period 2,...
- whereas the true relative price is $1 + r_1$.
- Only if the government sets $\tau_1 = \tau_2$, does this intertemporal distortion disappear.

4. Failure of Ricardian Equivalence: Tax Cuts and Twin Deficits

Distortionary Taxation: Firms

- ▶ For firms, same as in the economy with lump-sum taxes.
- ▶ The investment and profit schedules are given by (5) and (6), and therefore they are independent of the consumption tax rates τ_1 and τ_2 .

4. Failure of Ricardian Equivalence: Tax Cuts and Twin Deficits

Distortionary Taxation: Equilibrium

- ▶ Under free capital mobility, $r_1 = r^*$.
- ▶ Solve the Euler equation (27) for C_2 .
- ▶ Eliminate C_2 from the household's intertemporal budget constraint (26).
- ▶ Replace r_1 with r^* .
- ▶ This yields the equilibrium level of C_1 :

$$C_1 = \frac{1}{2(1 + \tau_1)} \left((1 + r_0)B_0^h + Q_1 + \frac{\Pi(r^*)}{1 + r^*} \right). \quad (28)$$

- ▶ In equilibrium C_1 is decreasing in τ_1 . An increase in τ_1 makes C_1 more expensive, inducing households to cut demand.
- ▶ $\rightsquigarrow$ Departure from Ricardian equivalence: given the path of government spending, changes in the timing of distortionary taxes have an effect on the equilibrium level of consumption.

4. Failure of Ricardian Equivalence: Tax Cuts and Twin Deficits

Distortionary Taxation: Equilibrium

- ▶ Consumption tax cut leads to current account deficit:
- ▶ $TB_1 = Q_1 - G_1 - I_1 - C_1$.
- ▶ Cut in τ_1 , holding constant G_1 , deteriorates TB_1 through the increase in C_1 :

$$\Delta TB_1 = -\Delta C_1 < 0$$

- ▶ The current account is equal to $CA_1 = r_0 B_0 + TB_1$.
- ▶ It also deteriorates and by the same magnitude as the trade balance:

$$\Delta CA_1 = \Delta TB_1 = -\Delta C_1 < 0.$$

- ▶ This shows that when taxes are distortionary, a cut in period-1 taxes can lead to an increase in the current account deficit.
- ▶ To establish whether the tax cut causes twin deficits, it remains to show that the tax cut, holding government expenditures constant, leads to a fiscal deficit.

4. Failure of Ricardian Equivalence: Tax Cuts and Twin Deficits

Distortionary Taxation: Equilibrium

- ▶ Consumption tax cut also leads to fiscal deficit:
- ▶ Tax revenue in period 1 is $\tau_1 C_1$.
- ▶ Government saving in period 1:

$$S_1^g = r_0 B_0^g + \tau_1 C_1 - G_1. \quad (29)$$

- ▶ The tax cut $\searrow$ revenue because $\tau_1 \searrow$ but it also $\nearrow$ revenue because the tax base $C_1 \nearrow$.
- ▶ Which effect dominates?
- ▶ Use equation (28) to eliminate C_1 in (29):

$$S_1^g = r_0 B_0^g + \frac{\tau_1}{2(1 + \tau_1)} \left((1 + r_0) B_0^h + Q_1 + \frac{\Pi(r^*)}{1 + r^*} \right) - G_1.$$

- ▶ The factor $\tau_1/(1 + \tau_1)$ is strictly $\nearrow$ in τ_1 .
- ▶ Therefore, $\tau_1 \searrow \rightsquigarrow$ government saving $\searrow \rightsquigarrow$ fiscal deficit $\nearrow$.
- ▶ We have therefore established that with distortionary consumption taxes, tax cuts can generate twin deficits.

5. The Optimality of Twin Deficits (not covered in class)

- ▶ With distortionary consumption taxes, tax cuts can generate twin deficits.
- ▶ But why would the government want to keep τ_1 constant when G_1 increases and finance the spending increase entirely with an increase in τ_2 ?
- ▶ Note that given (G_1, G_2) , there is an infinite number of tax rate paths, (τ_1, τ_2) , that guarantee the satisfaction of the government's intertemporal budget constraint and are consistent with equilibrium.
- ▶ Each of these tax rate paths gives rise to a different consumption path, (C_1, C_2) , and as a result generates a different level of welfare for the household.
- ▶ A natural question therefore is which of these tax rate paths a benevolent government should choose. The objective of this section is to address this question.

5. The Optimality of Twin Deficits (not covered in class)

Ramsey Optimal Consumption Tax Policy

- ▶ A *benevolent government* is a government that implements policies that maximize the welfare of households.
- ▶ The equilibrium tax rate path that maximizes the welfare of households is called the *Ramsey optimal* tax policy.
- ▶ (Frank Ramsey 1903-1930, British philosopher, mathematician, and economist)
- ▶ How to find the Ramsey optimal tax path, (τ_1, τ_2) ?
- ▶ Let's start by stating the equilibrium conditions.
- ▶ For simplicity, let's assume that $B_0^g = B_0^h = 0$.

5. The Optimality of Twin Deficits (not covered in class)

Ramsey Optimal Consumption Tax Policy

- ▶ Given the path of government spending (G_1, G_2) , an equilibrium is a path of tax rates (τ_1, τ_2) and a path of private consumption (C_1, C_2) , that satisfy :
 1. the household's Euler equation (27),
 2. and intertemporal budget constraint (26), both evaluated at $r_1 = r^*$,
 3. and the following intertemporal government budget constraint:

$$G_1 + \frac{G_2}{1 + r^*} = \tau_1 C_1 + \frac{\tau_2 C_2}{1 + r^*} \quad (30)$$

- ▶ These are three equations in four unknowns, $(C_1, C_2, \tau_1, \tau_2) \rightsquigarrow$ there are in principle many choices for the consumption tax path.
- ▶ The government can pick one of the two tax rates arbitrarily.
- ▶ This choice has welfare consequences for the household, because, in general, it affects the path of consumption.
- ▶ The question we wish to address is how should the government use this degree of freedom to maximize the household's well-being, or, in the words of Ramsey, "to minimize the decrement of utility."

5. The Optimality of Twin Deficits (not covered in class)

Ramsey Optimal Consumption Tax Policy: The Ramsey Problem

$$\max_{C_1, C_2, \tau_1, \tau_2} \ln C_1 + \ln C_2 \quad (7)$$

subject to

$$\frac{C_1}{C_2} = \frac{1 + \tau_1}{1 + \tau_2}(1 + r^*), \quad (31)$$

$$(1 + \tau_1)C_1 + \frac{(1 + \tau_2)C_2}{1 + r^*} = Q_1 + \frac{\Pi(1 + r^*)}{1 + r^*} \quad (32)$$

$$G_1 + \frac{G_2}{1 + r^*} = \tau_1 C_1 + \frac{\tau_2 C_2}{1 + r^*} \quad (30)$$

- ▶ The Ramsey problem seems daunting: three constraints and four control variables.
- ▶ However, it is a fairly easy problem to solve.

5. The Optimality of Twin Deficits (not covered in class)

Solving the Ramsey problem

- Combine equilibrium conditions (32) and (30), to obtain

$$C_1 + \frac{C_2}{1+r^*} = Q_1 - G_1 + \frac{\Pi(1+r^*) - G_2}{1+r^*} \quad (33)$$

- Note that equations (31), (32), and (30) are satisfied if and only if equations (31), (32), and (33) are satisfied.
- We can restate the Ramsey problem as picking $(C_1, C_2, \tau_1, \tau_2)$ to maximize the utility function (7) subject to (31), (32), and (33).
- Trick that makes solving this problem easy:
 - × Consider solving the less restricted problem of picking C_1 and C_2 to maximize the utility function (7) subject to (33) only.
 - × This problem has to deliver at least the same level of utility as the Ramsey problem, because it contains fewer restrictions.
 - × If we can show that the solution to the less restricted problem satisfies the omitted restrictions (31) and (32), we have found the solution to the Ramsey problem.

5. The Optimality of Twin Deficits (not covered in class)

The Less Restricted Ramsey Problem

- Solve (33) for C_2 :

$$C_2 = (1 + r^*)(\bar{Y} - C_1), \quad (34)$$

where $\bar{Y} = Q_1 - G_1 + \frac{\Pi(1+r^*)-G_2}{1+r^*}$

- Use (34) to eliminate C_2 from the utility function (7).
- The less restricted Ramsey problem then is

$$\max_{C_1} \ln C_1 + \ln \left((1 + r^*)(\bar{Y} - C_1) \right).$$

- The first-order condition is

$$\frac{1}{C_1} - \frac{1}{\bar{Y} - C_1} = 0. \quad (35)$$

- Solving for C_1 yields

$$C_1 = \frac{1}{2} \bar{Y} \quad (36)$$

5. The Optimality of Twin Deficits (not covered in class)

The Less Restricted Ramsey Problem

- Using (34) to eliminate $\bar{Y} - C_1$ from (35) gives

$$\frac{C_2}{C_1} = 1 + r^* \quad (37)$$

so that $C_2 = \frac{1+r^*}{2} \bar{Y}$

- This completes the solution of the less restricted Ramsey problem.
- Because $\bar{Y}$ does not depend on τ_1 and τ_2 , C_1 and C_2 do not depend on τ_1 and τ_2 .

5. The Optimality of Twin Deficits (not covered in class)

The Less Restricted Ramsey Problem

- ▶ Let's now show that the solution to the less restricted problem also solves the Ramsey problem.
- ▶ It is the case of it satisfies restrictions (31), (32), and (33).
- ▶ Equation (33) is trivially satisfied, as it is the constraint of the less restricted problem.
- ▶ Now pick $\tau_1 = \tau_2 = \tau$.
- ▶ Then, restriction (31) collapses to equation (37) from the less restricted problem.
- ▶ Finally, replacing τ_1 and τ_2 by τ in equation (32) gives the value of τ that makes this equation hold

$$\tau = \frac{Q_1 + \frac{\Pi(r^*)}{1+r^*}}{Q_1 - G_1 + \frac{\Pi(r^*) - G_2}{1+r^*}} - 1 \quad (38)$$

- ▶ This completes the proof that the solution of the less restricted problem is indeed the solution of the Ramsey problem.

5. The Optimality of Twin Deficits (not covered in class)

The Ramsey Optimal Taxation Problem: taking stock

1. Equation (37) says that the optimal tax policy completely eliminates the distortions introduced by the consumption tax. This is because under the Ramsey optimal tax policy, the intertemporal price of consumption perceived by the household, $\frac{1+\tau_1}{1+\tau_2}(1+r^*)$, is equal to the intertemporal price of consumption in the world market, $1+r^*$.
2. The way in which the benevolent government achieves a nondistorted path of consumption is by *tax smoothing*; that is, by setting a constant consumption tax rate over time, $\tau_1 = \tau_2$.
3. The Ramsey optimal allocation is the same as under lump-sum taxes. That is, the household consumes one half of the economy's lifetime resources net of government spending, $\bar{Y}$, in period 1 and leaves the rest for consumption in period 2.

5. The Optimality of Twin Deficits (not covered in class)

Are Twin Deficit Optimal?

- Let's compute the current account.

$$CA_1 = Q_1 - C_1 - I(r^*) - G_1.$$

- Replacing C_1 by its Ramsey optimal value $\bar{Y}/2$, we have that

$$CA_1 = \frac{1}{2} \left(Q_1 - G_1 + \frac{\Pi(r^*) - G_2}{1 + r^*} \right) + I(r^*). \quad (21)$$

which is the same as under lump-sum taxation (see equation (21) on slide 14).

(Recall that here we are assuming that $B_0^g = B_0^h = 0$)

- Public savings is

$$S_1^g = \tau_1 C_1 - G_1.$$

- Replacing τ_1 and C_1 by their Ramsey optimal values:

$$S_1^g = \frac{1}{2} \left(-G_1 + \frac{G_2}{1 + r^*} \right)$$

5. The Optimality of Twin Deficits (not covered in class)

Are Twin Deficits Optimal?

- ▶ Let's now consider a $\Delta G_1 > 0$ and $\Delta G_2 = 0$.
- ▶ The optimal way to finance this increase in G_1 is to keep $\tau_1 = \tau_2 = \tau$ and choose the new tax rate according to (28):

$$\tau + \Delta\tau = \frac{Q_1 + \frac{\Pi(r^*)}{1+r^*}}{Q_1 - (G_1 + \Delta G_1) + \frac{\Pi(r^*) - G_2}{1+r^*}} - 1$$

- ▶ Then we have that

$$\underbrace{\Delta CA_1}_{\text{Current Account deficit}} = -\frac{1}{2}\Delta G_1 < 0 = \underbrace{\Delta S_1^g}_{\text{Fiscal deficit}}.$$

- ▶ This means that the benevolent government finds it optimal to run twin deficits.

5. The Optimality of Twin Deficits (not covered in class)

Intuition for the Optimality of Twin Deficits

► Why a current account deficit?

- × An increase in G_1 makes households poorer.
- × By the income effect households cut consumption in both periods making the fall in C_1 less than the increase in government spending.
- × Since investment is constant at $I(r^*)$, the fact that consumption falls by less than government spending increases means that aggregate demand goes up.
- × With the endowment unchanged, the increase in aggregate demand causes a deterioration in the current account.

► Why a fiscal deficit?

- × In response to the increase in government spending in period 1, the government wants to avoid distortions in the intertemporal allocation of consumption.
- × it increases the consumption tax rate in both periods by the same amount.
- × Thus, in period 1 only part of the increase in government spending is financed with higher taxes and the rest by issuing public debt.

6. Fiscal Policy in Economies with Imperfect Capital Mobility

- ▶ A common concern about expansionary fiscal policy is that by driving up interest rates it crowds out investment and private consumption.
- ▶ Thus far, this effect was absent because by free capital mobility and the fact that the economy is small so that $r_1 = r^*$.
- ▶ We now relax these assumptions in a small open economy with different degrees of capital mobility

6. Fiscal Policy in Economies with Imperfect Capital Mobility

Modelling Imperfect Capital Mobility

- ▶ Assume that
 - × $r_1 = r^*$ if the country is a net external creditor ($B_1 > 0$),
 - × $r_1 > r^*$ and is an increasing function of the country's net debt position, B_1 , if the country is a debtor ($B_1 < 0$).
- ▶ Why?
 - × External indebtedness may increase the risk of default and therefore command a risk-premium.
 - × The government may impose capital controls to discourage external indebtedness.
- ▶ Assume that $B_0 = 0$. Then $B_1 = CA_1$.
- ▶ Then we can write imperfect capital mobility as:

$$r = \rho(-CA_1) = \begin{cases} r^* & \text{if } CA_1 \geq 0 \\ > r^* \text{ with } \rho' > 0 & \text{if } CA_1 < 0 \end{cases}$$

6. Fiscal Policy in Economies with Imperfect Capital Mobility

- ▶ Consider the effects of a change in government spending in period 1 financed with consumption taxes τ_2 (τ_1 is unchanged).
- ▶ Current account schedule:

$$CA_1 = r_0 B_0 + Q_1 - C_1 - I_1 - G_1.$$

- ▶ Using (28) evaluated at r_1 to eliminate C_1 and recognizing that I_1 is a function of r_1 alone yields the CA schedule:

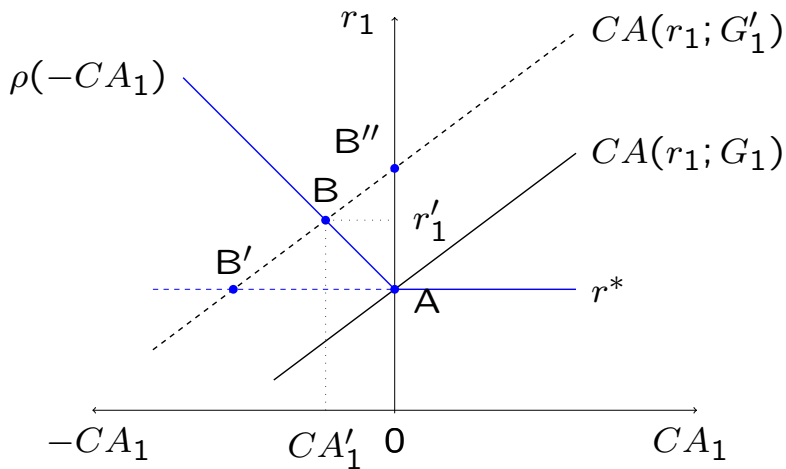
$$CA_1 = r_0 B_0 + Q_1 - \frac{1}{2(1 + \tau_1)} \left((1 + r_0) B_0^h + Q_1 + \frac{\Pi(r_1)}{1 + r_1} \right) - I(r_1) - G_1$$

which we write compactly as

$$CA_1 = CA(\underset{+}{r_1}; \underset{+}{Q_1}, \underset{+}{\tau_1}, \underset{-}{G_1}). \quad (39)$$

6. Fiscal Policy in Economies with Imperfect Capital Mobility

Adjustment to an Increase in Government Spending under Imperfect Capital Mobility



6. Fiscal Policy in Economies with Imperfect Capital Mobility

Discussion of the figure from the previous slide

- ▶ We assume that before increasing G_1 , $r_1 = r^*$ regardless of the degree of capital mobility.
- ▶ Increasing G_1 shifts the CA_1 schedule to the North West.
- ▶ B' corresponds to perfect capital mobility ($r_1 = r^*$)
- ▶ B'' corresponds to financial autarky
- ▶ B corresponds to imperfect capital mobility
- ▶ The more open to international capital mobility the economy is, the smaller the crowding out of investment, the smaller the increase in the interest rate, and the larger the deterioration of the current account following an expansion in government spending will be.

7. Summing Up

- a. This chapter analyzed how fiscal deficits stemming from tax cuts or increases in government spending affect the current account and other macroeconomic indicators.
- b. The twin deficit hypothesis states that fiscal deficits cause current account deficits.
- c. When taxes are lump sum, changes in the timing of taxes, holding the path of government spending unchanged, do not cause changes in the trade balance or the current account. This result is known as Ricardian equivalence.
- d. Ricardian equivalence is a fragile result. It requires that taxes are lump sum, that households are not borrowing constrained, and that all agents receiving a tax cut expect to pay the future higher taxes required to balance the budget.
- e. If any of these conditions is relaxed, Ricardian equivalence fails and tax cuts cause current account deficits (twin deficits).

7. Summing Up

- f. A temporary increase in government spending causes a deterioration of the current account (twin deficits).
- g. If an increase in government spending is perceived to be permanent, the current account is not significantly affected.
- h. A cut in the consumption tax rate, holding the path of government spending constant, causes a deterioration of the current account (twin deficits).
- i. A benevolent government finds it desirable to smooth consumption tax rates over time. Consequently, temporary increases in government spending generate both fiscal and current account deficits. In this sense, twin deficits are Ramsey optimal.
- j. In a small open economy with free capital mobility, temporary increases in government expenditures or tax cuts do not crowd out investment. However, crowding out of investment does occur in small open economies with imperfect international capital mobility.

